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Kinetic analysis of push-up exercises: a systematic review with practical recommendations

Wissem Dhahbi[a,b,c], Helmi Chaabene[d], Anis Chaouachi[a,e,f], Johnny Padulo[g,h], David G Behm[i], Jodie Cochrane[j], Angus Burnett[j,k] and Karim Chamari[k]

[a]Tunisian Research Laboratory, Sport Performance Optimization, National Center of Medicine and Science in Sports, Tunis, Tunisia; [b]Training Department, Qatar Police College, Doha, Qatar; [c]Sport Science Program, College of Arts and Sciences, University of Qatar, Doha, Qatar; [d]Division of Training and Movement Sciences, Research Focus Cognition Sciences, University of Potsdam, Germany; [e]Sports Performance Research Institute, AUT University, Auckland, New Zealand; [f]PVF Football Academy, Hang Yen, Vietnam; [g]Sport Performance Laboratory, Faculty of Kinesiology, University of Split, Split, Croatia; [h]Department of Psychology, University of eCampus, Novedate, Italy; [i]School of Human Kinetics and Recreation, Memorial University of Newfoundland, Newfoundland, Canada; [j]Centre for Exercise and Sport Science Research, School of Medical and Health Sciences, Joondalup Campus, Edith Cowan University, Perth, Australia; [k]Athlete Health and Performance Research Centre, Aspetar, QATAR Orthopaedic and Sports Medicine Hospital, Doha, Qatar

ABSTRACT

Push-ups represent one of the simplest and most popular strengthening exercise. The aim of this study was to systematically review and critically appraise the literature on the kinetics-related characteristics of different types of push-ups, with the objective of optimising training prescription and exercise-related load. A systematic search was conducted in the electronic databases PubMed, Google Scholar and Science Direct up to April 2018. Studies that reported kinetic data (e.g. initial and peak-force supported by the upper-limbs, impact-force, peak-flexion-moment of the elbow-joint, rate of propulsive- and impact-, and vertebral-joint compressive-forces) related to push-ups and included trained, recreational and untrained participants, were considered. The risk of bias in the included studies was assessed using the Critical Appraisal Skills Programme scale. From 5290 articles retrieved in the initial search, only 26 studies were included in this review. Kinetic data for 46 push-up variants were assessed. A limitation of the current review is that the relationship between our findings and actual clinical or practical consequences is not statistically proven but can only be inferred from our critical descriptive approach. Overall, this review provides detailed data on specific characteristics and intensities of push-up variations, in order to optimise exercise prescription for training and rehabilitation purposes.

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KEYWORDS
Biomechanics; closed kinetic chain exercise; ground reaction force; joint load; upper body

Introduction

Push-up exercises are widely used either for upper-extremity strengthening or for rehabilitation purposes (Faigenbaum et al., 2015; Ludewig, Hoff, Osowski, Meschke, & Rundquist, 2004; Moseley, Jobe, Pink, Perry, & Tibone, 1992). The popularity of push-up

CONTACT Wissem Dhahbi wissem.dhahbi@gmail.com
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exercises may be attributed to the fact that they have several technical variations and can be performed with little or no equipment. Additionally, the skills needed to perform push-ups are simple to master and the exercise intensity can easily be adjusted (Cogley et al., 2005) and adapted to various difficulty levels (Gouvali & Boudolos, 2005; Lou, Lin, Chou, Chou, & Su, 2001; Mier, Amasay, Capehart, & Garner, 2014). Push-ups are commonly used for dynamic warm-up, as an alternative to traditional weight training (Ubinger, Prentice, & Guskiewicz, 1999; Vossen, Kramer, Burke, & Vossen, 2000). In addition, push-ups are commonly used to rehabilitate the muscles of the upper extremities, particularly the shoulders (Ludewig et al., 2004). Moreover, push-up-based exercises can be used in performance assessments of strength endurance (ACSM, 2013) for school-age children (Loyd, Bishop, Walker, Sharp, & Richardson, 2003), military recruits (Dhahbi et al., 2017a; Dhahbi, Chaouachi, Padulo, Behm, & Chamari, 2015; Knapik et al., 2006) and elite athletes (Baker & Newton, 2006; Roe et al., 2015).

Recent studies have shown that a short-term (i.e. eight weeks) push-ups-based training programme improves upper-limb maximum dynamic strength in college baseball players (Carter, Kaminski, Douex, Knight, & Richards, 2007) and muscle-power performance in healthy college-aged participants (Schulte-Edelmann, Davies, Kernozek, & Gerberding, 2005). In the same context, practising push-up exercises has been shown to improve sport-specific performance (Klnç, 2008). For instance, Klnç (2008) reported significant shooting performance improvement after 10 weeks of a push-ups-based training programme in women basketball players aged 20 years. The same authors attributed this improvement to strength and power performance enhancement following the push-ups-based training programme. Additionally, it has been demonstrated that six weeks of push-ups practice decreases shear forces, improves proprioception and dynamic stability (Swanik et al., 2002) and enhances co-contraction and muscle recruitment (Rogol, Ernst, & Perrin, 1998; Ubinger et al., 1999). In terms of acute effects (i.e. post-activation potentiation), previous studies have proved that push-up exercises improve neuromuscular coordination (Ludewig et al., 2004) and increase joint compressive forces and stability (Calatayud, Borreani, Colado, Martin, & Rogers, 2014).

Several studies have analysed muscles' level of activation during different types of push-ups, to distinguish their respective intensities (Beach, Howarth, & Callaghan, 2008; Freeman, Karpowicz, Gray, & McGill, 2006; Gouvali & Boudolos, 2005). However, studies concerning the kinetics-based quantification of push-up exercises are scarce. The standard push-up (i.e. hands underneath shoulders, straight legs and body in a straight line) is the most popular exercise (Suprak, Dawes, & Stephenson, 2011). It includes many variations through simple changes in hand and/or leg positions. These variations affect the upper-body muscle activation (McGill, Cannon, & Andersen, 2014). Thus, it could be feasible to design a whole strength-training programme for the upper body using the different push-up variations (Ubinger et al., 1999; Vossen et al., 2000). Despite the popularity of push-ups, there is no previous study that reviews the literature on the kinetic characteristics of various push-up exercises. Therefore, a summary and critical review of the literature on the kinetic data for push-ups is needed, to improve our understanding and assist in determining and adjusting the exercise load.

The kinetic parameters of push-ups have been obtained using two distinct estimation methods: (i) *ground reaction forces (GRFs)* using a force platform to directly quantify the load applied on the arms (Dhahbi et al., 2017a; Dumas, Cheze, & Frossard, 2009) and (ii)

inverse dynamics (Bisseling & Hof, 2006). The inverse dynamics method calculates joint kinetics, (i.e. forces, moments and power) using direct measurement of the *GRFs* from force platforms or force transducers and calculates joint position data from motion analysis (Dumas et al., 2009). By estimating the mass and inertial properties of a segment, the joint kinetics can be determined (Bisseling & Hof, 2006). An, Korinek, Kilpela, and Edis (1990) and Donkers, An, Chao, and Morrey (1993) were the first researchers to analyse the kinetic and kinematic characteristics of different push-up exercises. They developed and proposed a method for collecting kinematic and kinetic data of the elbow joint during push-up exercises. The same authors provided information on elbow-joint loading patterns (An et al., 1990; Donkers et al., 1993).

Push-ups are an effective strength exercise that can be easily used and integrated into a training programme without any extra tools (Cogley et al., 2005; Dhahbi et al., 2017a). Therefore, the aim of this study was to comprehensively review and critically appraise previous research on the kinetic analysis of push-up exercises. Such a study may allow coaches to better understand the kinetic characteristics of the various types of push-up exercise and to make informed decisions regarding their training prescriptions.

Methods

Search strategy

The present systematic review of the published literature was conducted based on the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)* guidelines (Moher, Liberati, Tetzlaff, & Altman, 2010). A comprehensive literature search of original manuscripts was systematically performed on PubMed (MEDLINE), Google Scholar, and Science Direct. The search was limited to manuscripts published up to April 2018. A Boolean search strategy was applied using the operators 'AND' and 'OR'. According to the main topic of the present study, different combinations of the following terms were included in our search syntax: [('push-up' OR 'pushup') AND ('kinetic analysis' OR 'dynamic analysis') AND ('ground reaction force' OR 'inverse dynamic') AND ('joint loading' OR 'reaction force' OR 'joint compressive force' OR 'body weight' OR 'body mass' OR 'pressure') AND ('intensity' OR 'load')]. Additionally, the snowballing technique was applied to the reference lists of retrieved full-text articles to identify further articles that were not included in the initial electronic search. Only full-text sources were included. Thus, abstracts and conference papers from annual meetings were excluded from the analysis.

Eligibility criteria

The criteria for the inclusion of retrieved articles were that they: (i) were written in English, (ii) were published in peer-reviewed journals, (iii) addressed one or more push-up exercise types (iv) considered a targeted population which contained non-pathological participants (i.e. trained, recreational and untrained), (v) used a kinetic analysis approach and (vi) reported at least one aspect of either intensity and supported load or joint compressive load assessment.

Data extraction

Relevant articles identified through the searching process were independently evaluated and assessed by three reviewers (i.e. WD, DB and JC) who screened the titles, the abstracts and the full texts to reach a final decision on the study's eligibility. In case of uncertainty or disagreement, a fourth expert was consulted.

For each eligible study, data were extracted and examined by two independent reviewers (WD and JC) who performed the data extraction following a predefined template. The template included cohort size, demographic information including sex, age and training background/expertise, push-up variation name and the objective and design of the study. After completion of data extraction, the two independent reviewers cross-checked the accuracy of the data. Any conflicting results between reviewers resulted in a re-evaluation of the paper in question until a consensus was reached.

Assessment of study quality

The *critical appraisal skills programme (CASP)* score criteria of the Oxford Centre for Evidence-based Medicine were used for methodological assessment of the quality of the included cohort studies (Zeng et al., 2015). The quality evaluation of included studies was performed by two investigators. Any disagreement in study selection or quality assessment was resolved by further discussion.

Results

The preliminary systematic search resulted in 5290 hits. After careful examination of titles and abstracts, 407 articles remained and were subsequently reviewed for eligibility. The full texts of these 407 articles were screened with regard to the previously defined inclusion/exclusion criteria. Finally, 26 articles were retained and included in this study (Figure 1). The *CASP* scores for assessing the methodological quality of the 26 cohort studies are shown in Table 1. Table 2 illustrates the main characteristics of the included studies in terms of kinetic data for the respective push-up exercise, measured outcomes and the number and sex of participants, as well as their training background.

Publication years ranged from 1990 (An et al., 1990) to 2018 (Giancotti, Fusco, Varalda, Capranica, & Cortis, 2018). There are many variations of push-ups, as they can vary according to factors such as starting position, the *range of motion (ROM)* and movement speed. In total, we were able to identify46 push-up variations, which were then grouped into subsets. The main push-up types addressed in the included studies were standard push-ups (14 studies), flexed-knee push-ups, inter-hand-width push-ups or depth/*ROM* push-ups, hands- or feet-elevated push-ups and plyometric push-ups (seven studies each) and pushing speed (one study) (Table 2).

Our literature review showed that studies were based on the following kinetic indices for characterising and analysing push-up exercises: initial force supported by upper-limbs, peak force supported by upper-limbs, peak flexion moment of elbow joint, impact force, rate of force development (e.g. propulsive and impact) and vertebral-joint compressive forces. The kinetic indices were quantified using two distinct estimation methods, that is direct measurement of *GRFs* and inverse dynamics. The main findings of the analysed studies showed that irrespective of the push-up variant, the

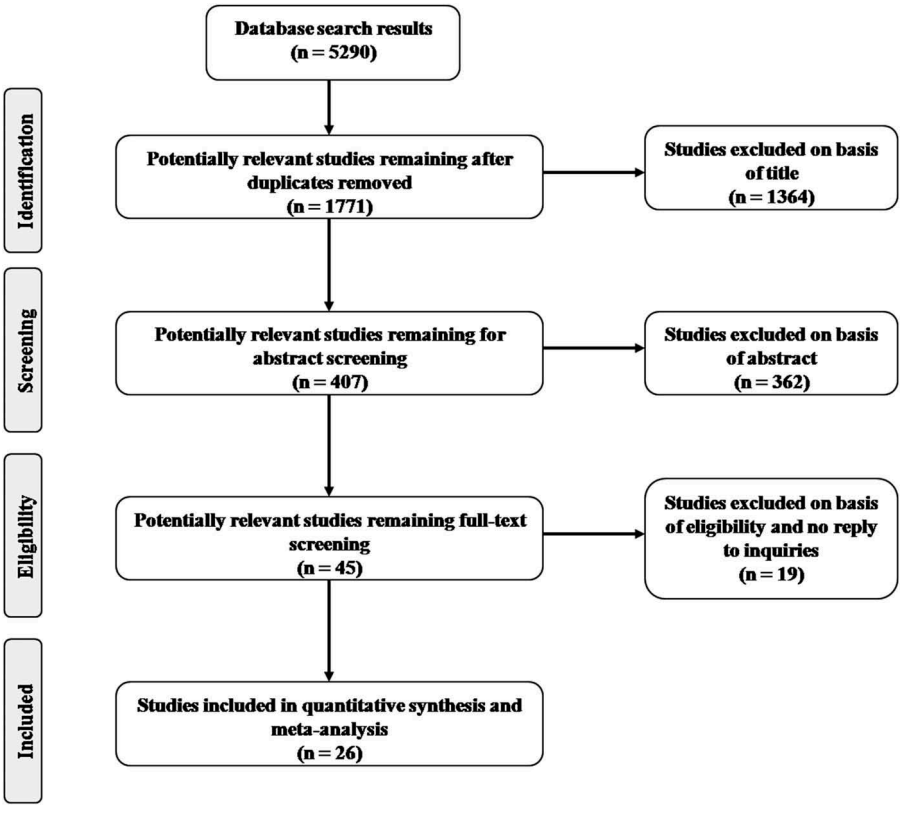


Figure 1. Flow of information through the different phases of the systematic review.

load supported by the arms with the elbows flexed is significantly greater (8–21%) than when the elbows are extended (Giancotti et al., 2018; Gulmez, 2016; Mier et al., 2014; San Juan, Suprak, Roach, & Lyda, 2015; Suprak et al., 2011). The force supported by both extended limbs, measured directly, ranged from 19% of *body weight (BW)* for suspended (e.g. TRX) 45° push-ups (Gulmez, 2016) to 73% *BW* for posterior/inferior push-ups (Gouvali & Boudolos, 2005) (Table 4). The force supported by the extended dominant limb, measured directly, ranged from 6% *BW* for 'prayer' push-ups (Uhl, Carver, Mattacola, Mair, & Nitz, 2003) to 60% *BW* for one-arm push-ups (Uhl et al., 2003) (Table 4). The initial axial force on the elbow for extended dominant-limb variants ranged from 21% *BW* for 'fast' push-ups (Chou et al., 2011) to 41% *BW* for standard push-ups (An et al., 1990) (Table 4). The peak *GRF* for both limbs ranged from 29% *BW* for anterior push-ups (Gouvali & Boudolos, 2005) to 131% *BW* for standard countermovement push-ups (Hogarth, Deakin, & Sinclair, 2013) (Table 5). The peak *GRF* for dominant-limb variants ranged from 67% *BW* for 3.8-cm 'box-drop' push-ups (Koch, Riemann, & Davies, 2012) to 78% *BW* for 'clap' push-ups (Moore, Tankovich, Riemann, & Davies, 2012) (Table 5). The peak axial force on the elbow for dominant-limb variants ranged from 33% *BW* for internal-hand-position push-ups (Lou et al., 2001) to 46% *BW* for arms-together (i.e. adducted arms) push-ups (Donkers et al., 1993) (Table 5). For plyometric push-ups, the impact force for both

Table 1. The critical appraisal skill program (*CASP*) score for assessing the methodological quality for the 26 enrolled clinical cohort studies.

Authors	*CASP* 1	*CASP* 2	*CASP* 3	*CASP* 4	*CASP* 5	*CASP* 6	*CASP* 7	*CASP* 8	*CASP* 9	*CASP* 10	*CASP* 11	*CASP* 12
Hinshaw et al. (2018)	+	+	+	+	+	+	+	+	+	+	+	+
Giancotti et al. (2018)	+	+	+	+	+	+	+	+	+	+	+	+
Polovinets et al. (2017)	+	+	+	+	?	+	+	+	+	+	+	+
Wang et al. (2017)	+	+	+	+	+	+	+	+	+	+	+	+
Dhahbi et al. (2017a)	+	+	+	+	+	+	+	+	+	+	+	+
Gulmez (2016)	+	+	+	+	?	+	+	+	+	+	+	+
San Juan et al. (2015)	+	+	+	+	+	+	+	+	+	+	+	+
McGill et al. (2014)	+	+	+	+	+	+	+	+	+	+	+	+
Mier et al. (2014)	+	+	+	+	+	+	+	+	+	+	+	+
Hogarth et al. (2013)	+	+	+	+	+	+	+	+	+	+	+	+
Koch et al. (2012)	+	+	+	+	+	+	+	+	+	+	+	+
Moore et al. (2012)	+	+	+	+	+	+	+	+	+	+	+	+
Garcia-Masso et al. (2011)	+	+	+	+	+	+	+	+	+	+	+	+
Suprak et al. (2011)	+	+	+	+	+	+	+	+	+	+	+	+
Ebben et al. (2011)	+	+	+	+	+	+	+	+	+	+	+	+
Chou et al. (2011)	+	+	+	+	?	+	+	+	+	+	?	+
Chuckpaiwong and Harnroongroj (2009)	+	+	+	+	+	+	+	+	+	+	+	+
Beach et al. (2008)	+	+	+	+	-	+	+	+	+	+	+	+
Chou et al. (2008)	+	+	+	+	-	+	+	+	+	+	?	+
Gouvali and Boudolos (2005)	+	+	+	+	+	+	+	+	+	+	+	+
Uhl et al. (2003)	+	+	+	+	+	+	+	+	+	+	+	+
Lou et al. (2001)	+	+	+	+	+	+	+	+	+	+	?	+
Hrysomallis and Kidgell (2001)	+	+	+	+	+	-	+	+	+	+	+	?
Donkers et al. (1993)	+	+	+	+	+	?	+	+	+	+	+	+
An et al. (1992)	+	+	+	+	+	?	?	+	+	+	+	+
An et al. (1990)	+	+	+	+	+	+	+	+	+	+	?	+

The eligibility criteria has to be excluded for scoring of the *CASP* score; '+' = indicates a 'yes' score; '–' = indicates a 'no' score; '?' = indicates a 'can't tell' score.

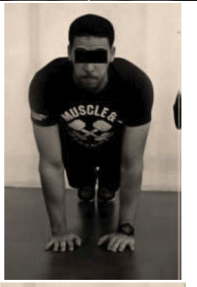
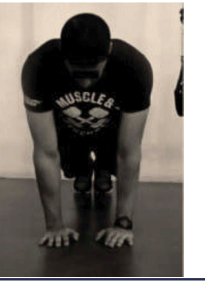
limbs ranged from 65% *BW* for 'jump' push-ups (Garcia-Masso et al., 2011) to 192% *BW* for standard squat push-ups (Dhahbi et al., 2017a). The propulsion rate force for both limbs ranged from 0.32 *BW*/s for kneeling squat push-ups (Dhahbi et al., 2017a) to 8 *BW*/s for standard countermovement push-ups (Hogarth et al., 2013). The propulsion rate force for dominant limb variants ranged from 2.9 *BW*/s for 3.8-cm box-drop push-ups (Koch et al., 2012) to 3.3 *BW*/s for clap push-ups (Koch et al., 2012). The rate of impact force for both limbs ranged from 17 *BW*/s for jump push-ups (Garcia-Masso et al., 2011) to 49 *BW*/s for standard countermovement push-ups (Dhahbi et al., 2017a). The rate of impact force for dominant-limb variants ranged from 4 *BW*/s for 3.8-c m box-drop push-ups(Koch et al., 2012) to 7 *BW*/s for clap push-ups (Koch et al., 2012). The peak flexion moment for dominant-limb variants ranged from 12 Nm for 'apart' variants (Donkers et al., 1993) to 29 Nm for 'together' variants (Donkers et al., 1993) (Figure 2). The vertebral joint compressive forces ranged from 1233 N for standard push-ups (McGill et al., 2014) to 3400 N for suspended push-ups (Beach et al., 2008). Additionally, there is a controversy about the dominant/non-dominant effects on push-up kinetic parameters. Regardless of push-up style, the *initial force supported by the upper-limbs* (F_0) (Polovinets, Wolf, & Wollstein, 2017) and the lateral and medial elbow-joint force in the 'up' position, are greater on the dominant limb than on the

Table 2. Variations of push-up exercise.

Cited by	Variation	Figure
Standard push-ups		
An et al. (1990); An et al. (1992); Beach et al. (2008); Chou et al. (2008); Chuckpaiwong and Harnroongroj (2009); Ebben et al. (2011); Gouvali and Boudolos (2005); Donkers et al. (1993); Hogarth et al. (2013); Lou et al. (2001); Mier et al. (2014); San Juan et al. (2015); Uhl et al. (2003)	**Standard**: The participant's hand placement is defined as the width equal to the distance between contralateral acromion processes, measured from the inside border of each hand, with hands placed under the shoulders in the beginning position, which is characterised by full elbow extension.	
Pushing speed		
Chou et al. (2011)	**Fast-speed push-up**:7 push-ups/10s. **Regular-speed push-up**:5 push-ups/10s. **Slow-speed push-up**:4 push-ups/10s. Push-up: The participant keeps the elbow joints in full extension and position their hands, with a forearm in an axially non-rotated posture. The hand width is set to 1.5 times of the shoulder width and the feet are shoulder-width apart in the initial posture.	
Knee angle		
Gouvali and Boudolos (2005); Ebben et al. (2011); Mier et al. (2014); San Juan et al. (2015); Suprak et al. (2011)	**Flexed knee (knees down)**: The participant kneels on the floor on hands and knees, then walks the hands forward until they can drop the hips. They stop when the body is in a straight line from the shoulders to the knees. Wrists are lined up beneath your shoulders. The participant crosses the ankles, with their feet in the air. Knee push-ups are performed in the same manner as the standard push-up except they are on their knees instead of their feet.	
Uhl et al. (2003)	**Prayer**: The participant kneels with weight shifted primarily over the ankles and leans forward. The prayer position is illustrated to represent a standing upper extremity weight-bearing exercise in which minimal weight is placed through the upper extremities. Note that the participant places their middle fingers along two taped lines on the scales; these are used to standardise hand position based on individual biacromial widths.	

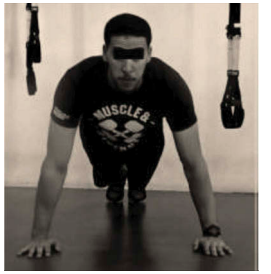
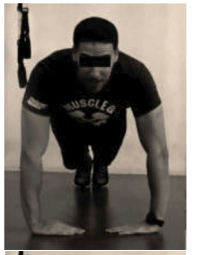
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Table 2. (Continued).

Cited by	Variation	Figure
Uhl et al. (2003)	**Quadruped**: The participant is positioned with hands and knees on the ground with the shoulder flexed to 90°.	
Variation of hand position (width) push-ups		
An et al. (1992); Chou et al. (2008); Donkers et al. (1993); Gouvali and Boudolos (2005)	**Shoulder abduction**: Two marks for the position of the hands at the abducted variant, at a distance of ~150% of the shoulder width.	
An et al. (1992); Donkers et al. (1993); Chou et al. (2008); Gouvali and Boudolos (2005)	**Shoulder adduction**: Two marks for the position of the hands at the adducted variant, at a distance of ~50% of the shoulder width.	
Chuckpaiwong and Harnroongroj (2009)	**Push-up -10 (P-10)**: Hand interval~10 cm narrower than the shoulder width on each side.	

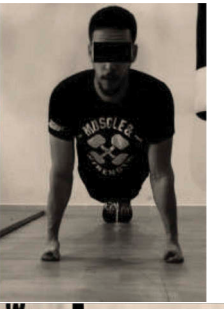
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Table 2. (Continued).

Cited by	Variation	Figure
Chuckpaiwong and Harnroongroj (2009)	**Push-up + 10, + 20, or + 30 (P + 10, P + 20 or P + 30)**: Hand interval -10, 20 or 30 cm wider than the shoulder width on each side.	
Depth/*ROM* push-ups		
Lou et al. (2001)	**Internal position**: Starting from the standard push-up position, with arms at the participant's sides, and palms facing forward as a neutral reference position. The hands are internally rotated at 90°.	
Lou et al. (2001)	**External position**: Starting from the standard push-up position, with arms at the participant's sides, and palms facing forward as a neutral reference position. The hands are externally rotated at 90°.	

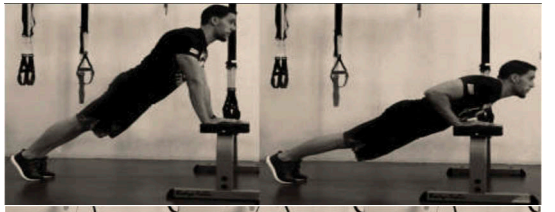
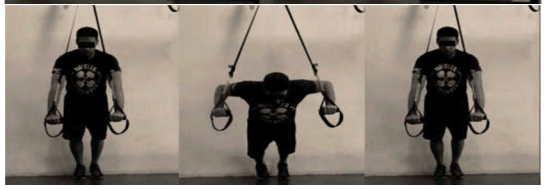
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Table 2. (Continued).

Cited by	Variation	Figure
Polovinets et al. (2017)	**Neutral wrist**: Starting from the standard push-up position, but with neutral rotation of the wrist and forearm, thus the knuckles were placed along the body of the participant.	
Gouvali and Boudolos (2005)	**PV**: Two marks for the position of the feet at the anterior variant, at a distance of + 30% of the arm-forearm length, relative to the marks of the feet in the normal position.	
Gouvali and Boudolos (2005)	**AV**: Two marks for the position of the feet at the posterior variant, at a distance of −30% of the arm-forearm length, relative to the marks of the feet in the normal position.	
An et al. (1992); Chou et al. (2008); Donkers et al. (1993)	**Superior**: The hands are placed 15 cm in the cephalad direction from the standard push-up position.	

(*Continued*)

Table 2. (Continued).

Cited by	Variation	Figure
An et al. (1992); Chou et al. (2008); Donkers et al. (1993)	**Inferior**: The hands are placed 15 cm in the caudal direction from the standard push-up position	
Hand or feet-elevated push-ups		
Ebben et al. (2011); Lehman et al. (2006); Uhl et al. (2003)	**Push-up position with feet elevated on a 30, 45 or 61 cm box (FE30, FE61or FE45)**:Push-up feet elevated. The participant maintains a push-up position with feet elevated at … cm, with elbows in full extension and the shoulder flexed to 90°.	
Ebben et al. (2011); Lehman et al. (2006); Uhl et al. (2003)	**Hands elevated on a 30 or 61 cm box (HE30 or HE61)**: Push-up hands elevated. The participant maintains a push-up position with hands elevated at … cm, with elbows in full extension and the shoulder flexed to 90°.	
Gulmez (2016); McGill et al. (2014); Snarr and Esco (2013)	**TRX® push-up-with** the TRX straps hanging vertically. The participant adopts a push-up position with a handle in either hand. They perform a push-up in the same manner and at the same pace as the standard push-up.	
Plyometric push-ups		
Koch et al. (2012); Moore et al. (2012)	**Clap push-ups**: The participant lowers their chest down towards the force plates, while maintaining a straight line with their body. After this phase, they forcefully and explosively push-up in order to obtain an upper body take-off. While in the air, they clap their hands together before landing back onto the force plates (or ground).	

(*Continued*)

Table 2. (Continued).

Cited by	Variation	Figure
Koch et al. (2012); Moore et al. (2012)	**Box-drop plyometric push-ups3.8, 7.6 or 11.4 cm (3.8BD, 7.6BD or 11.4BD)**: This position is similar to the clap push-ups. The participant lowers their chest down towards the force plates, while maintaining their body in a straight line. Once peak depth is obtained, they explosively push up and, while in the air, manoeuvre their hands, such that they land on the blocks simultaneously. Upon contact with the blocks, the participant immediately and explosively performs another push-up off the blocks, landing with their hands in the starting position (on the force plates) to perform a subsequent repetition.	
Garcia-Masso et al. (2011); Hrysomallis and Kidgell (2001)	**Countermovement push-up/explosive push-up**: This is a push-up performed with a maximum speed of movement. From the starting position (standard position), the participant descends by flexing their elbows and extending their arms until the sternum is approximately 3 cm from the floor. They immediately extend their elbows and flex their arms to return to the starting position.	
Garcia-Masso et al. (2011)	**Jump push-up**: This is similar to a clap push-up but without the clap. From the starting position (standard position), the participant descends by flexing their elbows and extending their arms until the sternum is approximately 3 cm from the floor. They immediately extend their arms by taking advantage of the inertia generated to take their hands off the floor. They are told not to flex their elbows during the flight phase. Upon landing, the participant must reduce the impact as much as possible and continue the movement, thereby carrying out another complete push-up with a flight phase and landing. During the measurement, the participant must carry out two complete repetitions of the exercise (i.e. eccentric phase, concentric phase, flight phase and landing phase) performed at maximal speed.	

(*Continued*)

Table 2. (Continued).

Cited by	Variation	Figure
Garcia-Masso et al. (2011)	**Fall push-up**: *Starting position*: The participant kneels with bent arms (90°), extended elbows and palms facing forward. *Execution*: The participant lets himself/herself fall forward and, at the moment of impact, tries to minimise the impact-force as much as possible. They must place their hands on the pre-set marks. They lower their bodies until the sternum is approximately 3 cm from the floor. They then extend their elbows and flex their arms, thereby taking advantage of the inertia generated by these movements, so as to take their hands off the floor and attempt to return to the starting position.	
Dhahbi et al. (2017a)	**Standard countermovement push-up (SCPU)**: From the 'up' standard position, the participant descends by flexing their elbows until the elbows are flexed at approximately 90°. They immediately extend their arms to propel the upper body as high as possible. They are requested not to flex their elbows during the flight phase. Upon landing, the participant is requested to reduce the impact as much as possible and resume the starting position.	
Dhahbi et al. (2017a)	**Standard squat push-up (SSPU)**: From the 'up' standard position, the participant descends by flexing their elbows until the elbows are at an angle of approximately 90°, and they maintain this position for about 1 s. When their body is completely stabilised in this motionless position, they then propel their upper body as high as possible without performing a countermovement.	
Dhahbi et al. (2017a)	**Kneeling countermovement push-up (KCPU)**: The KCPU is performed in the same manner as the SCPU, except with support from the knees instead of the feet (i.e. kneeling position).	

(*Continued*)

Table 2. (Continued).

Cited by	Variation	Figure
Dhahbi et al. (2017a)	**Kneeling squat push-up (KSPU)**: The KSPU is performed in the same manner as the SSPU, except with support from the knees instead of the feet.	
Dhahbi et al. (2017a)	**Drop-fall push-up**: The participant kneels with shoulders in antepulsion (90°), with elbows extended and palms facing forward. The participant then lets himself/herself fall forward (under the effect of gravity). They lower their body until the elbows are flexed at approximately 90°. They immediately extend their arms in the same manner as the KCPU.	
Various types		
Donkers et al. (1993); Uhl et al. (2003)	**One-arm push-up**: The participant maintains a one-arm push-up position with the elbow in full extension, the dominant shoulder flexed to 90° and the non-dominant hand placed behind the back during testing. The participant is also asked to perform a one-handed push-up if able to do so.	
Uhl et al. (2003)	**Tripod**: The participant remains in the quadruped position, then flexes the non-performing shoulder to 180°	
Uhl et al. (2003)	**Pointer**: The participant maintains the same position as the tripod position and extends the contralateral hip to 0°.	

non-dominant limb (Lou et al., 2001). When performing plyometric push-ups, the dominant upper-limb side demonstrated significantly greater peak forces and rate of impact force (1.5% and 37% respectively) compared to the non-dominant side (Koch et al., 2012). For all suspended push-up variations, no difference in F_0 was found between dominant and non-dominant arms (Gulmez, 2016). Moreover, there is a controversy about the sex effects on push-up kinetic parameters. Mier et al. (2014) found that F_0 on the flexed elbow, (i.e. the 'down' position) as well as the peak force supported by the upper-limbs (PF) are significantly greater for men than for women. In the same context, Hinshaw, Stephenson, Sha, and Dai (2018) also found a significant effect of sex on F_0 and PF. However, the same authors did not establish pairwise comparisons between men and women. Conversely, for both flexed-knee and standard push-ups, Mier et al. (2014) did not find any significant difference between men and women with regard to F_0 during the extended elbow (i.e. up) position. Similarly, Ebben et al. (2011) did not report any effect of sex on *PF* when performing different push-up variations (i.e. standard, flexed-knee, feet-elevated and hands-elevated push-ups). Note that regardless of push-up type, the neutral hand has been shown to be at the less-stressful position for wrist and elbow joints (Chuckpaiwong & Harnroongroj, 2009; Lou et al., 2001; Polovinets et al., 2017).

Discussion

Standard push-ups

The standard push-up is the most extensively studied (An, Chao, Morrey, & Donkers, 1992; An et al., 1990; Beach et al., 2008; Chou et al., 2008; Chuckpaiwong & Harnroongroj, 2009; Donkers et al., 1993; Ebben et al., 2011; Gouvali & Boudolos, 2005; Hogarth et al., 2013; Lou et al., 2001; Mier et al., 2014; Polovinets et al., 2017; San Juan et al., 2015; Uhl et al., 2003) (Table 2).

Initial force as a percentage of body weight supported by upper-limbs via direct measurement (F_0)

The load supported by the arms in the course of the push-up, is largely determined by the participant's *BW* and how it is supported on the ground (Mier et al., 2014) (Table 4). Some authors have only assessed the dominant limb (the one used for writing (Polovinets et al., 2017)) for kicking (Garcia-Masso et al., 2011) or for throwing (Uhl et al., 2003)), while others have evaluated both limbs together (Dhahbi et al., 2017a; Gouvali & Boudolos, 2005; Hogarth et al., 2013; Mier et al., 2014; San Juan et al., 2015; Suprak et al., 2011) or have used both methods (Gulmez, 2016). Gouvali and Boudolos (2005) recorded the force patterns during six different push-ups and found that a clear distinction does exist in *GRF* between the different postures adopted for each push-up type.

Overall, 75% *BW* was supported by the upper-limbs during a standard push-up when the elbows were in flexion (Mier et al., 2014), while *GRF*s reached 69% *BW* when the elbows were extended during the standard push-up (Freeman et al., 2006; Gouvali & Boudolos, 2005; Mier et al., 2014; Suprak et al., 2011). However, Gouvali and Boudolos (2005) demonstrated that the posterior push-up had the highest level of *GRF* supported by both arms (i.e. 73% *BW*). This information can help coaches to choose an

appropriate intensity-related type of push-up when developing and guiding training programmes.

Initial elbow joint axial force (as a percentage of BW, estimated with inverse dynamics—$F_{0'}$)

Many researchers have assessed the initial joint axial forces (as a percentage of *BW*) on the elbow (i.e. $F_{0'}$). These forces are observed to be heterogeneous for different push-up variations, but they are similar across studies on the same variant, with the exception of studies by Chou et al. (2011) and Lou et al. (2001). The latter studies reported lower values for a standard push-up, with an axial joint force on the elbow of 28% *BW*, with other results showing values from 37% to 41% *BW* (An et al., 1992, 1990; Donkers et al., 1993). This may be explained by the mesomorphic body type and the lower distribution of mass in the upper body of the Taiwanese cohort included in the Chou et al. (2011) and Lou et al. (2001) studies (Table 3). It may also be related to the calculation models used in different laboratories. According to Lou et al. (2001), the heterogeneity in the anthropometric data for tested participants (e.g. mass and inertia of segments) could result in a 5% difference in the force calculations.

The three-segment model used in Lou et al. (2001) assumed that the hand and forearm behaved as two different rigid bodies. In contrast, Donkers' model (Donkers et al., 1993) assumed the hand and forearm to be one rigid body. This difference may contribute to differences in force and moment calculations. The greater the number of segments in the model, the greater the measurement error rate (Dumas et al., 2009; Lou et al., 2001). Observing Table 4, it can be concluded that the $F_{0'}$ values were approximately similar to the results from the direct measurement method for the same push-up variation. It is important to consider the elbow-joint forces when designing a training programme. However, it should be acknowledged that the elbow-joint force varies, depending on the body positioning and elbow angle, which affect the moment arm and torques.

Peak force relative to BW supported by upper-limbs (PF)

The present review of the literature revealed that for the standard push-up, the values of the *PF* for both arms exhibited some variation ranging from 41% to 98% *BW*. This large range of values may be due to the different measurement approaches used in different investigations. For instance, Gouvali and Boudolos (2005) zeroed the force plate when the participant's hands were on the force plate (initial position of push-up) and found a *PF* of41% *BW*, whilst other researchers, such as Mier et al. (2014), zeroed the force plate when it was free of any weight and reported a *PF* of 98% *BW*. Additionally, it has been shown that the performing velocity of push-ups can affect results (e.g. *PF* and *PF'*) (Chou et al., 2011; Mier et al., 2014). In this regard, Mier et al. (2014)instructed participants to perform one push-up in two seconds, whereas in the study of Gouvali and Boudolos (2005), push-ups were performed at the participant's own pace.

Peak elbow-joint axial forces (as a percentage of BW, estimated with inverse dynamics—PF')

Four studies (An et al., 1990; Chou et al., 2011; Donkers et al., 1993; Lou et al., 2001) have examined the standard push-up and measured the *PF'* of the dominant limb. The

Table 3. Kinetic studies of push-up variations.

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Hinshaw et al. (2018)	-18 male and 17 female physically active - 21.9 ± 3.5 y - 70.2 ± 13.5 kg - 1.74 ± 0.10 m	- Modified plyometric push-up on the knees (MP) - Plyometric push-up without external loading (PP) - Plyometric push-up with an external load of 5% body weight (PP5%) - Plyometric push-up with an external load of 10% body weight (PP10%)	- **Force Applied to Hands at Starting Position (*BW*)**: For men and women: MP< PP< PP5%< PP10% - **Force Applied to Feet at Starting Position (*BW*)**: For men and women: PP< PP5%< PP10%< MP - **Peak Force Applied to Hands during Push-up (*BW*)**: For men: MP< PP and PP5%< PP10% For women: MP< PP< PP5%< PP10% - **Peak Upper Body Power during Push-up (watt/*BW*)**: For men and women: MP< PP,PP5% and PP10% - **Peak Velocity during Push-up (m/s)**: For men: MP< PP10%< PP5%< PP For women: MP< PP5% and PP10%< PP - **Force at Peak Power during Push-up (*BW*)**: For men and women: MP< PP< PP5%< PP10% - **Velocity at Peak Power during push-up (m/s)**: For men: MP< PP10%< PP5%< PP For women: MP< PP5% and PP10%< PP
Giancotti et al. (2018)	17 male and 8 female physically active - 28.1 ± 5.2 y - 69.4 ± 14.3 kg - 171.6 ± 11.3 cm	Suspension push-up at different length of suspension training device 178 cm (P1), 188 cm (P2), 198 cm (P3), 208 cm (P4), 218 cm (P5), 228 cm (P6) and 238 cm (P7)	**For the cell load measurement**: ICC: 0.91–0.96 **For the force plate measurement**: ICC: 0.97–0.98 - **Load supported by hands (*BW*)**: for both flex and extend elbow positions; P1< P2< P3< P4< P5< P6< P7 - **Load supported by feet (*BW*)**: for both flex and extend elbow positions; P7< P6< P5< P4< P3< P2< P1
Polovinets et al. (2017)	-14 healthy male - 24.86 ± 0.95 y - 176.09 ± 3.06 cm, - 22.94 ± 1.42 kg· m^{-2}	- Hyperextended wrist push-up (HW) - Neutral wrist push-up (NW)	- All calculations of ICC showed good agreement between tests (ICC:0.69–0.96) - The total *GRF* measured did not differ between the 2 patterns of push-ups. - The upper-limbs were loaded with an average of 70% of total body weight during the experience. - The applied *GRF* on the dominant hand was higher regardless of push-up style. - When performing NW, the *GRF* vector was more uniform throughout the push-up in the vertical direction. - The horizontal distance between the capitate bone location and the *GRF* origin was smaller in hyperextension. - The forces travelled more dorsally over a wider area and more ulnarly in the hyperextended wrist.

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Table 3. (Continued).

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Wang et al. (2017)	-60 recreationally active men - 24.5 ± 4.3 y - 1.75 ± 0.07 m - 80.8 ± 13.5 kg	Ballistic push-up (i.e. Standard countermovement push-up)	**Peak Force (N)**: ICC: 0.971; SEM: 45 **Mean Force (N)**: ICC: 0.989; SEM: 21 **Peak *RFD* (N/s)**: ICC: 0.849; SEM: 1059 **Mean *RFD* (N/s)**: ICC: 0.867; SEM: 469 **Net Impulse (N· s)**: ICC: 0.956; SEM: 5.9 **Peak Velocity (m/s)**: ICC: 0.863; SEM: 0.12 **Flight Time (s): ICC**: 0.750; SEM: 0.05 **Peak Power (W): ICC**: 0.936; SEM: 123 **Mean Power (W): ICC**: 0.934; SEM: 66
Dhahbi et al. (2017a)	-37 commandos physically active and used to doing push-ups - 23.3 ± 1.5 y - 78.7 ± 9.7 kg - 1.80 ± 0.04 m	- Standard countermovement push-up (SCPU) - Standard squat push-up (SSPU) - Kneeling countermovement push-up (KCPU) - Kneeling squat push-up (KSPU) - Drop-fall push-up (DFPU)	- **Initial force supported (*BW*) and impact-force (*BW*)**: SCMJ-PU and SSJ-PU> KCMJ-PU, KSJ-PU and DJF-PU - **Peak ground reaction force take-off (*BW*)**: SCMJ-PU> DJF-PU> SSJ-PU and KCMJ-PU> KSJ-PU - **Rate of force development during take-off (*BW*/s)**: SCMJ-PU and DJF-PU> KCMJ-PU> SSJ-PU and KSJ-PU - **Flight time (s)**: DJF-PU> KCMJ-PU and KSJ-PU> SCMJ-PU and SSJ-PU - **Rate of force development impact (*BW*/s)**:No difference
Gulmez (2016)	-28 physically healthy males - 24.1 ± 2.9 y - 78.8 ± 9.8 kg - 179.4 ± 8.0 m	- TRX push-up angle 45° (P1) - TRX push-up angle 30° (P2) - TRX push-up angle 15° (P3) - TRX push-up angle 0°(P4)	- **Load applied to the TRX straps (kg)**: *P4> P3> P2> P1 *For all variations, elbow flexion> elbow extension *For all variations, no difference between dominant and non-dominant arms
San Juan et al. (2015)	- Total of 18 males and 4 females - 28.4 ± 10.1 y - 75.3 ± 10.3 kg - 176.9 ± 7.9 cm	- Traditional push-up plus (TPUP) - Modified push-up plus (MPUP)	- **Vertical *GRF* (%*BW*)**: *TPUP (+ 17% *BW*)> modified PUP *TPUP at 90° of elbow flexion (75.99%*BW*) *TPUP at 20° of elbow flexion (70.18%*BW*) *MPUP at 90° of elbow flexion (52.95% *BW*)> MPUP at 20° of elbow flexion (57.95%*BW*) *Vertical *GRF* displayed a significant linear decrease across the *ROM* for both variants
McGill et al. (2014)	-40 physically healthy males - 21.1 ± 2.0 y - 74.6 ± 7.8 kg, - 1.77 ± 0.06 m.	- Standard push-up - TRX push-up - TRX push-angle 1 - TRX push-angle 2 - TRX push-angle3 - Stable shoulder protraction (coached) - TRX scapula push-up - TRX shoulder protraction (coached) - TRX shoulder protraction (not coached) - Stable shoulder protraction (not coached)	- **Mean spine (L4/L5) compression at the P-phase of each exercise**: TRX push-angle3> TRX push-up> TRX push-angle 2> stable shoulder protraction (coached)> TRX scapula push-up> TRX shoulder protraction (coached)> TRX push-angle 1> standard push-up> TRX shoulder protraction (not coached)> stable shoulder protraction (not coached) -**Mean spine (L4/L5) shear forces at the P-phase of each exercise**: Standard push-up; TRX push-up> TRX push-angle 2> TRX push-angle3> TRX push-angle 1

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Table 3. (Continued).

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Mier et al. (2014)	19 physically active men: - 25 ± 7 y - 78.6 ± 10.7 kg - 1.77 ± 0.08 cm 18 physically active women: - 22 ± 3 y - 64.1 ± 6.8 kg - 166 ± 6 cm	- Dynamic standard push-up (DSP) - Dynamic modified push-up (DMP) - Static standard push-up 'up' - Static standard push-up 'down' - Static modified push-up 'up' - Static modified push-up 'down'	- **Maximal load supported during DSP and DMP (*BW*)**: Men> women - **Load supported in the static 'down' position (*BW*)**: Men> women - **Load supported in the static 'up' position (*BW*)**: No difference was observed between men and women - **Load supported in the static position (*BW*)**: Standard push-up> modified push-up
Hogarth et al. (2013)	-14 men physically active and used to doing push-ups - 19 ± 1y, - 92.6 ± 14.6 kg - 1.83 ± 0.07 m	- Plyometric push-up (i.e. SCPU)	- **Peak force (N), mean force (N), rate of force development during take-off (N/s), time to peak force (ms), flight time (ms), impulse (N· s) and initial mass supported (N)**: Moderate to high relative reliability (ICC = 0.8–0.96) and high absolute reliability (CV = 4.3–7.6%), except for rate of force development during take-off (CV = 11%)
Koch et al. (2012)	- 22 men physically active - 25.9 ± 1.3 y - 87.6 ± 12 kg - 1.80 ± 0.08 m	- Clap push-up (CPU) - 3.8 cm box-drop push-up (3.8BD) - 7.6 cm box-drop push-up (7.6BD) - 11.4 cm box-drop push-up (11.4BD)	- **Peak *GRF* (*BW*)**: *Dominant limb> non-dominant limb *No significant difference between variations in push-ups - **Impact rate (*BW*/s)**: *Dominant limb> non-dominant limb *Within dominant limb, CPU> 11.4BD> 7.6 BD> 3.8 BD - **Propulsion rate (*BW*/s)**: CPU> 3.8 BD, 7.6BD, 11.4BD
Moore et al. (2012)	- 21 recreationally active adult males - 24.5 ± 3.7 y - 83.2 ± 11.7 kg - 1.82 ± 0.05 m	- Clap push-up (CPU) - 3.8 cm box-drop push-up (3.8BD) - 7.6 cm box-drop push-up (7.6BD) - 11.4 cm box-drop push-up (11.4BD)	- **Peak *GRF* (%*BW*)**: CPU> 3.8 BD, 7.6BD, 11.4BD
Garcia-Masso et al. (2011)	- 27 men physically active and used to doing push-ups - 22.44 ± 0.31 y - 76.34 ± 0.97 kg - 1.79 ± 0.01 m	- Countermovement push-up (CMPU) - Jump push-up (JPU) - Fall push-up (FPU)	- **Maximum force (N)**: CMPU> FPU - **Rate of force development (N/s)**: CMPU> JPU - **Impact-force (N)**:FPU> JPU - **Rate of impact-force development (N/s)**: FPU> JPU
Suprak et al. (2011)	- 28 highly strength trained men - 33.6 ± 6.59 y - 84.66 ± 11.65 kg - 178.98 ± 5.76 cm	- Standard push-up 'up' (RU) - Standard push-up 'down' (RD) - Push-up flexed knee 'up' (FKU) - Push-up flexed knee 'down' (FKD)	- **Percentage of body mass supported (%*BW*)**: RD> RU> FKD> FKU - **Percentage change in body weight supported from up to down position in traditional and modified push-ups (%*BW*)**: - Push-up flexed knee> standard push-up

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Table 3. (Continued).

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Ebben et al. (2011)	- 14 men: - 22.5 ± 3.6 y - 83.70 ± 11.28 kg, 179.81 ± 10.89 cm - 9 women: - 21.1 ± 1.6 y - 69.34 ± 7.90 kg - 173.43 ± 9.37 cm	- Push-up with feet elevated on a 30.48/60.96 cm box - Push-up with hands elevated on a 30.48/60.96 cm box - Push-up with flexed knee (FK) - Standard push-up (R)	- Peak *GRF* expressed as a coefficient of total body mass: *FE 60.96> FE 30.48 > R> HE 30.48> HE 60.96> FK - Peak *GRF* (N): *FE 60.96, FE 30.48 > R, HE 30.48, HE 60.96, FK *No difference between men and women
Chou et al. (2011)	-11 physically healthy males - 24.54 ± 1.44 y - 65.9 ± 7.17 kg - 168.9 ± 5.32 cm	- Fast-speed push-up: 7 push-ups/10s (F) - Regular-speed push-up: 5 push-ups/10s (R) - Slow-speed push-up: 4 push-ups/10s (S)	**- Joint forces of elbow at the examined push-up speeds**: *Anterior/posterior force 'up' (%*BW*): RS> FS *Anterior/posterior force 'down' (%*BW*): SS> FS *Lateral/medial force 'up' (%*BW*):SS,FS> RS *Lateral/medial force 'down' (%*BW*):SS,RS> FS *Lateral/medial force 'peak' (%*BW*):RS,SS> FS *Axial force 'up', 'down' and 'peak' (%*BW*): RS,SS> FS **- Joint moments of elbow at the examined push-up speeds**: - Frontal plane 'peak' (N· m): RS,SS> FS - Sagittal plane 'down' and 'peak' (N· m): RS,SS> FS - Transverse plane 'down' and 'peak' (N· m): SS> FS
Chuckpaiwong and Harnroongroj (2009)	10 physically active men (20 hands) - 22.10 ± 0.7 y - 65.80 ± 7.5 kg - 1.73 ± 5.6 m	**- Push-up -10**:Hand interval 10 cm narrower than the shoulder width on each side (P-10) - Standard push-up (R) **- Push-up + 10, + 20, or + 30**: Hand interval -10, 20 or 30 cm wider than the shoulder width on each side (P + 10, P + 20 or P + 30)	**- Distribution of the mean peak pressure (N· cm^{-2})**: Lunate and hypothenar are as> thenar, metacarpal and finger are as in both 'up' and 'down' positions (descriptive data) **- Mean palmar peak pressure in lunate area during push-up in 'up' phase (N· cm^{-2})**: P + 30, P + 20 > P-10, R, P + 10 **- Mean palmar peak pressure in lunate area during push-up in 'down' phase (N· cm^{-2})**: P + 30 > P-10, R, P + 10, P + 20 **- Mean palmar peak pressure in hypothenar area during push-up in 'down' phase (N· cm^{-2})**: P-10, R > P + 10, P + 20, P + 30
Beach et al. (2008)	- 11 recreationally trained men - 27.4 ± 0.8 y - 89.4 ± 3.3 kg - 1.83 ± 0.02 m	- Standard push-up (R) - Suspended push-up (SP)	**- Mean and peak L4/L5 intervertebral joint compressive forces (N)**: S 'up'> SP 'down'> R 'up'> R 'down' **- Mean and peak L4/L5 intervertebral joint anterior/posterior shear forces (N)**: R 'up', SP 'up'> R 'down', SP 'down'
Chou et al. (2008)	- 10 healthy male students	- Standard push-up (R) - Apart push-up (A) - Shoulders' adduction push-up [50% shoulders' width] (AD) - Superior push-up (S) - Inferior push-up (I)	**- Maximum axial force (%*BW*)**: R > S,A - **Maximum flexion moment (N· m)**: AD> R > A

(*Continued*)

Table 3. (Continued).

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Gouvali and Boudolos (2005)	- 8 recreationally active adult males - 20.5 ± 0.4 y - 74.4 ± 5 kg - 176.8 ± 2.3 cm	- Standard push-up (R) - Shoulders' abduction [150% shoulders' width] (*AB*) - Shoulders' adduction [50% shoulders' width] (AD) - + 30% of arm-forearm length posteriorly to initial hand position (PV) - −30% of arm-forearm length anteriorly to initial hand position (AV) - Push-up flexed knee (FK)	- Significant differences existed for most vertical force variables (V*GRF* initial, V*GRF* initial/*BW*, V*GRF* max., V*GRF* max./*BW*, V*GRF* min., V*GRF* min./*BW*, V*GRF* integral during descending and V*GRF* range), but without presenting the results of *post hoc* tests - Not significant differences for anteroposterior force **- Initial load relative to body weight (%*BW*):** PV> R, *AB*, AD, AV> FK
Uhl et al. (2003)	- 18 healthy participants - 22 ± 3 y - 73 ± 17 kg - 175 ± 10 cm	- Prayer (P) - Quadruped (Q) - Tripod (T) - Pointer (Po) - Standard push-up (R) - Push-up with feet elevated on a 45 cm (FE45) - One-arm push-up (OA)	**- Percentage of body mass supported during 'up' position (%*BW*):** OA> T, Po, R, FE45 > Q > P
Lou et al. (2001)	- 10 physically healthy male - 26.1 ± 2.6 y - 69.3 ± 9.2 kg -171.7 ± 5.2 cm	- Standard push-up (R) - Internal position push-up (IR) - External position (ER)	**- Joint force of elbow at specific push-up events (%*BW*):** *Anterior/posterior force 'up': R,ER> IR *Anterior/posterior force 'peak': R> ER,IR *Lateral/medial force 'up': R,ER> IR (dominant hand), ER> R> IR (non-dominant hand) *Lateral/medial force 'peak': R,ER> IR *Axial force 'up': R,ER> IR (dominant hand) *Axial force 'down': R,ER> IR (non-dominant hand) *Axial force 'peak': R> ER,IR **- Joint force of elbow at specific push-up events (N· m):** *Frontal plane 'up': IR> ER> R (dominant hand), IR,ER> R (non-dominant hand) *Frontal plane 'down': IR,ER> R (non-dominant hand) *Sagittal plane 'up': R,ER> IR (dominant hand)
Hrysomallis and Kidgell (2001)	- 12 physically active men - 22.8 ± 3.0 y - 81.9 ± 9.2 kg - 181.3 ± 8.2 cm	- Explosive push-up	- No significant differences for any of the vertical force variables (impulse, maximum rate of force development, average force, peak force and initial mass on force plate) when the explosive push-ups were preceded by a set of 5RM bench press exercises

(*Continued*)

Table 3. (Continued).

Study	Participants (N, age, *BW*, height)	Push-ups tests	Significant findings ($p < 0.05$)
Donkers et al. (1993)	- 9 healthy male participant - 25 [20–30] y - 78 [65–90] kg - 178 [167–187] cm	- Standard push-up (R) - Repeated standard push-up (RR) - Apart push-up (A) - Shoulders' adduction push-up [50% shoulders' width] (AD) - Superior push-up (S) - Inferior push-up (I) - One-handed push-up (OH)	- Description of typical pattern of force and moment of push-up exercise. - **Static axial force 'up' (%*BW*)**: I > R> AD,S - **Maximum axial force (%*BW*)**: R > A, S - **Maximum anterior-posterior force (%*BW*)**: S > R > A - **Maximum flexion moment (N.cm)**: AD> R> RR,A - **Maximum valgus moment (N.cm)**: S > R
An et al. (1992)	- 9 healthy male participants - 25 [20–30] y.	- Standard push-up (R) - Repeated standard push-up (RR) - Apart push-up (A) - Shoulders' adduction push-up [50% shoulders' width] (AD) - Superior push-up (S) - Inferior push-up (I) - One-handed push-up (OH)	- Description of typical pattern of force and moment of push-up exercise - **Maximum axial force (%*BW*)**: R > A - **Maximum flexion moment (N cm)**: AD> R> RR,A
An et al. (1990)	- 1 male participant - 170 lbs 77.11 kg, - 6 ft 182.88 cm	- Standard push-up (R)	- 3D description of joint forces and moment of elbow - Presentation of typical pattern of 3D force and moment of push-up exercise

BW = body weight; *GRF* = ground reaction force; ROM = range of motion

Table 4. Initial load relative to body weight.

	Ratio of body weight supported by both limbs, with direct measurement				Ratio of body weight supported by dominant limb, with direct measurement				Initial axial force on elbow, relative to *BW*, for dominant limb			
Push-up group	**Variations**	**Studies**	**Values**	**SD**	**Variations**	**Studies**	**Values**	**SD**	**Variations**	**Studies**	**Values**	**SD**
Standard push-up	Standard	Suprak et al. (2011)	0.69	0.02	Standard	Uhl et al. (2003)	0.34	0.03	Standard	Chou et al. (2011)	0.28	0.04
	Standard	Gouvali and Boudolos (2005)	0.66	0.02					Standard	Donkers et al. (1993)	0.37	
	Standard	Mier et al. (2014) for men	0.67	0.04					Standard	An et al. (1990)	0.41	
	Standard	Mier et al. (2014) for women	0.65	0.03					Standard	An et al. (1990)	0.37	
	Standard	Dhahbi et al. (2017a)	0.68	0.04					Standard	Lou et al. (2001)	0.28	0.02
	Standard	Hogarth et al. (2013)	0.61	0.1					Standard	Polovinets et al. (2017)	0.35	0.02
	Standard for men	Hinshaw et al. (2018)	0.67	0.03								
	Standard for women	Hinshaw et al. (2018)	0.64	0.04								
	Standard	San Juan et al. (2015)	0.76						Neutral wrist	Polovinets et al. (2017)	0.35	0.03
Pushing speed									Fast	Chou et al. (2011)	0.21	0.03
									Slow	Chou et al. (2011)	0.29	0.04
Knee angle	Flexed knee	Suprak et al. (2011)	0.53	0.04	Prayer	Uhl et al. (2003)	0.06	0.03				
	Flexed knee	Gouvali and Boudolos (2005)	0.53	0.03	Quadruped	Uhl et al. (2003)	0.19	0.02				
	Flexed knee	Mier et al. (2014) for men	0.53	0.04								
	Flexed knee	Mier et al. (2014) for women	0.52	0.03								
	Flexed knee	Dhahbi et al. (2017a)	0.46	0.04								
	Flexed knee for men	Hinshaw et al. (2018)	0.52	0.05								
	Flexed knee for women	Hinshaw et al. (2018)	0.48	0.05								
	Flexed knee	San Juan et al. (2015)	0.53									

(*Continued*)

Table 4. (Continued).

	Ratio of body weight supported by both limbs, with direct measurement				Ratio of body weight supported by dominant limb, with direct measurement				Initial axial force on elbow, relative to *BW*, for dominant limb			
Push-up group	**Variations**	**Studies**	**Values**	**SD**	**Variations**	**Studies**	**Values**	**SD**	**Variations**	**Studies**	**Values**	**SD**
Inter-hand width, position and direction	Apart/shoulder abduction	Gouvali and Boudolos (2005)	0.63	0.03					Apart	Donkers et al. (1993)	0.37	
	Together/ shoulder adduction	Gouvali and Boudolos (2005)	0.65	0.02					Adduction	Donkers et al. (1993)	0.35	
	Posterior/inferior variation	Gouvali and Boudolos (2005)	0.73	0.03					Inferior	Donkers et al. (1993)	0.39	
	Anterior/superior variation	Gouvali and Boudolos (2005)	0.61	0.03					Superior	Donkers et al. (1993)	0.35	
									Internal position	Lou et al. (2001)	0.27	0.03
									External position	Lou et al. (2001)	0.29	0.03
Feet elevated/hands elevated	TRX 178 cm	Giancotti et al. (2018)	0.19	0.03	Push-up feet elevated at 45 cm	Uhl et al. (2003)	0.39	0.05				
	TRX 188 cm	Giancotti et al. (2018)	0.23	0.03								
	TRX 198 cm	Giancotti et al. (2018)	0.27	0.03	TRX 45°	Gulmez (2016)	0.07					
	TRX 208 cm	Giancotti et al. (2018)	0.32	0.03	TRX 30°	Gulmez (2016)	0.12					
	TRX 218 cm	Giancotti et al. (2018)	0.37	0.03	TRX 15°	Gulmez (2016)	0.18					
	TRX 228 cm	Giancotti et al. (2018)	0.40	0.02	TRX 0°	Gulmez (2016)	0.27					
	TRX 238 cm	Giancotti et al. (2018)	0.44	0.02								
	TRX 45°	Gulmez (2016)	0.13									
	TRX 30°	Gulmez (2016)	0.24									
	TRX 15°	Gulmez (2016)	0.37									
	TRX 0°	Gulmez (2016)	0.54									
Various stationary positions					Tripod	Uhl et al. (2003)	0.32	0.03				
					Pointer	Uhl et al. (2003)	0.34	0.04				
					One-arm push-up	Uhl et al. (2003)	0.60	0.06				

BW = bodyweight; SD = standard deviation

outcomes of these studies ranged from 36% to 45% *BW* (see Table 5). The results relative to the standard push-up *PF* established by An et al. (1990) and Donkers et al. (1993) seem to be high (45% of *BW*) compared to other studies (33% to 43% *BW*) (Chou et al., 2008; Lou et al., 2001). Donkers et al. (1993) found that a simple change of hand width can affect *PF*' by around 4% *BW*. Lou et al. (2001) showed that the standard position (i.e. hands facing forward as a neutral reference position) generated a significantly greater *PF*' than the 'internal' (i.e. hands internally rotated at 90°) and 'external' (i.e. hands externally rotated at 90°) hand positions.

Peak flexion moment of elbow joint (PFM)

Similar *PFM* outcomes have been observed (An et al., 1990; Chou et al., 2011; Donkers et al., 1993; Lou et al., 2001). The mean *PFM* value for the studies was around 23 Nm for the standard push-up (Figure 2).

Speed of push-up

Our literature search found that there was little information on the kinetic analysis of pushing speed on push-ups. Only Chou et al. (2011) investigated the kinetic implications (e.g. joint forces and moments) of performing push-ups at different speeds by performing seven push-ups in 10 s as a fast push-up, five push-ups in 10s as a regular-speed push-up and four push-ups in 10s as a 'slow' push-up.

F_0 and PF'

Following the analysis of Chou et al. (2011), the velocity of the push-up execution did not affect $F_{0'}$ (see Table 4). However, the push-up speed did influence *PF*', with a fast push-up producing a greater force by a factor of 1.2 than a slower push-up, at 43% and 35% of *BW* respectively (Chou et al., 2011) (Table 5). Additionally, it was observed by Chou et al. (2011) that performing push-ups at a slower speed reduces elbow-joint

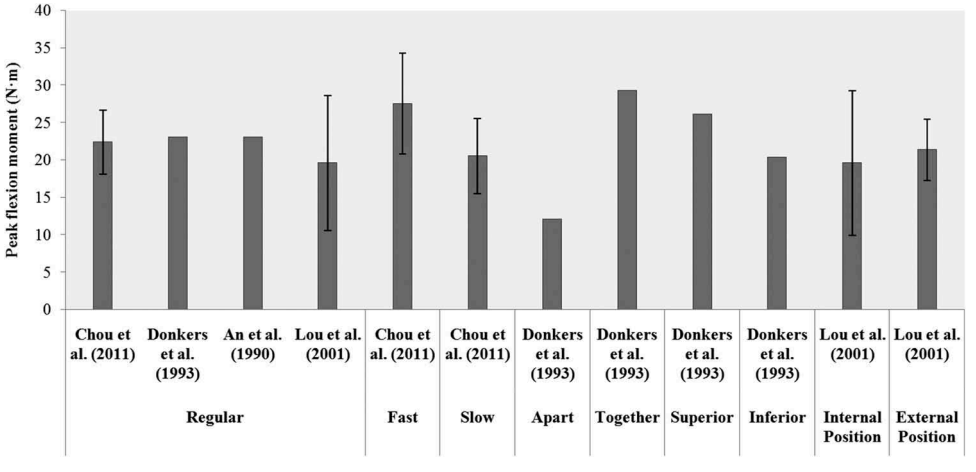


Figure 2. Peak flexion moment of joint elbow as an absolute value. *Values displayed with ± SD are in line with the availability in the literature.

Table 5. Peak force relative to body weight.

	Peak *GRF*, relative to body weight, for both limbs				Peak *GRF*, relative to body weight, for dominant limb				Peak axial force on elbow, relative to body weight, for dominant limb			
Push-up group	Variations	Studies	Values	SD	Variations	Studies	Values	SD	Variations	Studies	Values	SD
Standard push-up	Standard	Ebben et al. (2011)	0.64	0.04					Standard	Chou et al. (2011)	0.38	0.05
	Standard	Mier et al. (2014) for men	0.98	0.08					Standard	Donkers et al. (1993)	0.45	
	Standard	Mier et al. (2014) for women	0.80	0.04					Standard	An et al. (1990)	0.45	
	Standard	Gouvali and Boudolos (2005)	0.41	0.13					Standard	Lou et al. (2001)	0.36	0.04
Pushing speed									Fast	Chou et al. (2011)	0.43	0.06
									Slow	Chou et al. (2011)	0.35	0.05
Knee angle	Flexed knee	Ebben et al. (2011)	0.49	0.05								
	Flexed knee	Mier et al. (2014) for men	0.80	0.07								
	Flexed knee	Mier et al. (2014) for women	0.68	0.03								
	Flexed knee	Gouvali and Boudolos (2005)	0.35	0.01								
Inter-hand width, position and direction	Shoulder abduction	Gouvali and Boudolos (2005)	0.38	0.12					Apart	Chou et al. (2008)	0.43	
	Shoulder adduction	Gouvali and Boudolos (2005)	0.30	0.09					Apart	Donkers et al. (1993)	0.45	
	Posterior variation	Gouvali and Boudolos (2005)	0.34	0.08					Together	Donkers et al. (1993)	0.46	
	Anterior variation	Gouvali and Boudolos (2005)	0.29	0.07					Inferior variation	Donkers et al. (1993)	0.46	
									Superior variation	Chou et al. (2008)	0.44	
									Superior variation	Donkers et al. (1993)	0.42	
									Internal position	Lou et al. (2001)	0.33	0.03
									External position	Lou et al. (2001)	0.34	0.04
Plyometric push-up	Explosive push-up	Hrysomallis and Kidgell (2001)	0.67	0.18	Clap	Koch et al. (2012)	0.69	0.02				
	Explosive push-up	Garcia-Masso et al. (2011)	0.68	0.02	Clap	Moore et al. (2012)	0.78	0.15				
	Standard squat push-up	Dhahbi et al. (2017a)	1.04	0.10	3.8 cm box-drop	Koch et al. (2012)	0.67	0.02				
	Kneelings quat push-up	Dhahbi et al. (2017a)	0.82	0.10	3.8 cm box-drop	Moore et al. (2012)	0.69	0.11				

(*Continued*)

Table 5. (Continued).

	Peak *GRF*, relative to body weight, for both limbs				Peak *GRF*, relative to body weight, for dominant limb				Peak axial force on elbow, relative to body weight, for dominant limb			
Push-up group	Variations	Studies	Values	SD	Variations	Studies	Values	SD	Variations	Studies	Values	SD
	Standard countermovement push-up	Dhahbi et al. (2017a)	1.26	0.15	7.6 cm box-drop	Koch et al. (2012)	0.70	0.02				
	Standard countermovement push-up	Wang et al. (2017)	1.21	0.24	7.6 cm box-drop	Moore et al. (2012)	0.71	0.12				
	Standard countermovement push-up for men	Hinshaw et al. (2018)	1.29	0.16	11.4 cm box-drop	Koch et al. (2012)	0.69	0.02				
	Standard countermovement push-up For women	Hinshaw et al. (2018)	0.94	0.13	11.4 cm box-drop	Moore et al. (2012)	0.71	0.12				
	Standard countermovement push-up	Hogarth et al. (2013)	1.31	0.23								
	Kneeling countermovement push-up	Dhahbi et al. (2017a)	0.99	0.14								
	Kneeling countermovement push-up for men	Hinshaw et al. (2018)	1.07	0.16								
	Kneeling countermovement push-up for women	Hinshaw et al. (2018)	0.79	0.12								
	Kneeling countermovement push-up	Garcia-Masso et al. (2011)	0.69	0.02								
	Drop-fall push-up	Dhahbi et al. (2017a)	1.12	0.11								
	Fall (kneeling)	Garcia-Masso et al. (2011)	0.65	0.02								
Feet elevated/hands elevated	Feet elevated at 60.96 cm	Ebben et al. (2011)	0.74	0.02								
	Feet elevated at 30.48 cm	Ebben et al. (2011)	0.70	0.02								
	Hands elevated at 30.48 cm	Ebben et al. (2011)	0.55	0.05								
	Hands elevated at 60.96 cm	Ebben et al. (2011)	0.41	0.06								

BW = bodyweight; *GRF* = ground reaction force; SD = standard deviation

loading. Therefore, to lighten the load, performing a slow push-up may be recommended.

PFM

The fast and together variants are seen to be more stressful at the level of the elbow joint (with values higher than 27 Nm), whereas the apart (Donkers et al., 1993) variant produces the lowest intensity (12.11 Nm). Chou et al. (2011) showed that the *PFM*, the peak valgus moment and the pronation moment at fast push-up speeds are 1.6, 1.3 and 1.4 times greater respectively than at slow speeds. However, at lower speeds there is greater activation in the upper-limb and chest muscles. These observations indicate that performing low-speed push-ups is preferred for greater muscle activation and therefore a slow muscle contraction with sub-maximal resistance can substitute for stimulation with high-intensity resistance. For example, if the overall resistance is low relative to the strength potential of the athlete (such as a strong trainee who cannot access extra resistance and is performing standard body-weight push-ups), low-speed push-ups may increase muscle activation and therefore be more effective than faster contractions. In addition, performing low-speed push-ups seems to be ideal from an injury prevention perspective due to the fact that there is less torque on the elbow joints (Chou et al., 2011).

Flexed-knees push-up

F_0

The F_0 for the flexed-knee push-up was consistent across almost all studies (52–53% of *BW*) (Gouvali & Boudolos, 2005; Mier et al., 2014; San Juan et al., 2015; Suprak et al., 2011), except for one study which reported a slightly lower value of 46% *BW* (Dhahbi et al., 2017a). During the static kneeling push-up position, Suprak et al. (2011) observed a load of 53% of *BW* in the up (i.e. elbow extended) position compared to 62% of *BW* in the down (i.e. elbow flexed) position. Mier et al. (2014) investigated the effects of push-up variation (standard vs. kneeling) and sex on F_0. They demonstrated that the relative load did not differ between men and women in the static up position (Mier et al., 2014), whereas the relative load was higher in men than in women during the static down position (i.e. flexed elbow joint). These data demonstrated sex-related differences in the relative loads that resulted from the push-ups (Gouvali & Boudolos, 2005; Suprak et al., 2011). Women hold a slightly lower proportion of their skeletal muscle mass in the upper body (Janssen, Heymsfield, Wang, & Ross, 2000), as well as generally having a narrower chest and shoulder girth compared to men. This could contribute to a shorter moment arm in women and influence the torque and hence the loading and intensity of the push-up.

PF

Ebben et al. (2011) assessed the *PF* of push-up variations, including the standard push-up and those performed with a flexed knee. Push-ups from the flexed-knee position produced a lower *PF* than most push-up variations, with no sex-related differences (Table 5). Previous studies reported lower *PF* for kneeling than for standard push-ups. The three studies (Ebben et al., 2011; Gouvali & Boudolos, 2005; Mier et al., 2014) that measured *PF* in standard and kneeling push-ups showed a lower *PF* in the latter push-

up type. Specifically, the *PF* for the kneeling push-ups has been shown to be between 35% and 80% *BW*. On the other hand, the *PF* for the standard push-ups ranged from 41% to 98% *BW* (Ebben et al., 2011; Gouvali & Boudolos, 2005; Mier et al., 2014). Therefore, as the flexed-knee push-up showed a lower upper-body load, this would be an effective alternative for individuals who are unable to perform a standard push-up or who need a lighter load for any reason. This form of push-up is often chosen in order to promote optimal scapulothoracic stability in the early phases of weight-bearing shoulder rehabilitation standards (Suprak et al., 2011).

Inter-hand width (apart/shoulder abduction, together/shoulder adduction)

F_0 and $F_{0'}$

The initial force in inter-hand-width variations ranged from 61% to 73% *BW* (Dhahbi et al., 2017a; Ebben et al., 2011; Gouvali & Boudolos, 2005; San Juan et al., 2015; Suprak et al., 2011) for F_0 and from 35% to 39% *BW* for $F_{0'}$ (An et al., 1992, 1990; Chou et al., 2001, 2008; Donkers et al., 1993; Lou et al., 2001). These outcomes are similar to those for the standard push-up within the same study (Donkers et al., 1993; Gouvali & Boudolos, 2005) (see Table 4). It has been demonstrated that bearing a heavy weight on a *hyper-extended wrist (HW)* can cause early development of wrist pain and associated carpal tunnel syndrome (Gellman, Sie, & Waters, 1988). Polovinets et al. (2017) compared *in vivo GRF* load vectors during a *neutral wrist (NW)* push-up with those for push-ups performed on an *HW*. The main finding of his study showed that when push-ups are performed with the wrist and forearm in neutral positions, forces are transferred through the volar wrist and onto the broad articular surface of the distal radius through the radioscaphoid and radiolunate joints, therefore with less joint stress. Finally, Polovinets et al. (2017) recommended performing push-ups on an *NW* to prevent wrist ligament injury or pain.

Pf

The narrower hand position exhibited lower *PF* or axial elbow-joint force compared to the wider hand position (Donkers et al., 1993; Gouvali & Boudolos, 2005). The study of Chuckpaiwong and Harnroongroj (2009) revealed that a wider hand position generates a higher peak pressure on the medial side of the palm, whereas a narrower hand position induces a higher peak pressure on the lateral side of the palm. Therefore, a wider hand position may be the best choice when performing push-ups to prevent hypothenar pain, while a narrower hand position may be appropriate to prevent thenar or lunate pain (Table 5).

PFM

The together variants are more stressful at the level of the elbow joint (with values higher than 27 Nm), whereas the apart (Donkers et al., 1993) variant produces the lowest intensity (12.11 Nm). Donkers et al. (1993) showed that a considerably greater reduction in flexion torque resulted from hands being apart, while there was an increase in the *PFM* in the apart position when initiating the set of push-ups. Greater *PFM* torque existed throughout the cycle with hands together compared with the standard hand position.

Depth/range of motion ('posterior/inferior' variation, 'superior/anterior' variation, internally rotated hand position and externally rotated hand position)

F_0'

Lou et al. (2001) evaluated the elbow-joint forces during push-up exercises at various forearm rotations. These authors revealed that push-ups involving an internally rotated position should be avoided, in order to prevent excessive shear forces. Lou et al. (2001) also found that the externally rotated hand position and standard push-up position had significantly greater (although small) initial (up) axial force than the internally rotated position. Donkers et al. (1993) investigated the effects of five push-up variations (i.e. standard, apart, together, superior and inferior) on elbow-joint loading (Table 4).

PF'

The fast, apart, together, superior and inferior push-up variants displayed similar values. Moreover, these are the push-ups with the most intense peak axial force on the elbow (*PF*' > 43% *BW*), in comparison with other variants, such as internal position, external position and slow, where the force was lower (Table 5). Donkers et al. (1993) found that the maximum axial force decreased from the value for the standard position (45% *BW*) when hands were apart or superior, to 42.7% *BW* and 41.9% *BW* respectively. Lou et al. (2001) found that the standard position resulted in significantly greater forces than the internal and external variants. In general, the posterior/inferior variation imposed greater forces than the superior/anterior variation, and also greater forces than the standard push-up. Furthermore, the external position resulted in greater forces than the internal position, though they were similar to the standard push-up forces (Table 5).

Pfm

Significant differences in elbow-joint torques have been observed in studies for push-ups with varying hand positions. Donkers et al. (1993) found that the maximum valgus torque was significantly greater in a superior-hand-position push-up compared to the standard push-up. The principal findings of Lou et al. (2001) concern the frontal plane, with the valgus moment of the up-position event (i.e. extended elbow joint) during the standard push-up being greater than in the internal and external push-up variations. In addition, on the sagittal plane, the *PFM* of the up-position event was greatest in the internal position. Therefore, it would be logical to avoid a 90° internal rotation hand position in order to minimise mechanical stress on the elbow joint. Morrey, Tanaka, and An (1991) reported that the medial-collateral ligament is the major constraint for the elbow joint in resisting valgus stress, while the lateral-collateral ligament helps the elbow joint resist varus stress. These increases in joint loading increase the risk of injury to the medial and lateral stabilising structures of the elbow, putting the medial-collateral ligament, lateral-collateral ligament and flexor-pronator muscle group at potential risk of injury.

Hands- or feet-elevated push-ups

F_0

The elevated-hands exercise, as used by Ebben et al. (2011), involving the 61-cm box, is similar to the exercise described by Gulmez (2016). Ebben et al. (2011) showed that, for this position, where the elbows were in extension, the performer had an exercise load of 41% *BW*. The exercise load analysed by Gulmez (2016) for the same position was found to be 23% *BW* higher than that established by Ebben et al. (2011). This could be due to the unstable surface for TRX push-ups. Indeed, the values reported by Ebben et al. (2011) (relative loads of 64% *BW* and 49% *BW* for standard and kneeling push-ups respectively) were close to previous values (Gouvali & Boudolos, 2005; Mier et al., 2014; Suprak et al., 2011) observed for a static push-up position (Table 4).

Pf

Ebben et al. (2011) assessed the *PF* of push-up variations, including the standard push-up and those performed with a flexed knee, with feet elevated on a 30.48-cm box and a 60.96-cm box and with hands elevated on a 30.48-cm box and a 60.96-cm box. The main outcomes of this study proved that push-ups with the feet elevated produced a higher *PF* than all other push-up variations (Ebben et al., 2011). Furthermore, push-ups with hands elevated and push-ups from the flexed-knee position produced a lower *PF* than all other push-up variations, with no sex-related differences in these push-up variations (Ebben et al., 2011) (Table 5).

Plyometric push-ups

Pf

Dhahbi et al., (2017a) examined differences between *GRF*-based parameters collected from five types of plyometric push-up: standard countermovement push-up, standard squat push-up, kneeling countermovement push-up, kneeling squat push-up and 'drop-fall' push-up. They revealed that the *PF* during take-off was higher for standard countermovement push-ups compared to drop-fall push-ups, while the latter resulted in a higher *PF* than other exercises. These findings are in agreement with the findings of Garcia-Masso et al. (2011), who showed that *PF* during take-off of the countermovement push-up was higher than in the 'fall' push-up. This difference in *PF* may be due to the absence of a concentric phase during the descending movement. This suggests that the *PFs* of the performed countermovement push-up exercises are achieved at the start of the concentric phase of the propulsive phase. However, to confirm this hypothesis, future studies will have to use synchronised kinematic and kinetic analyses to dissociate concentric/eccentric contraction during the propulsion phase (Dhahbi et al., 2017a). Based on the study by Koch et al. (2012), the clap and box-drop push-ups from 3.8 cm (box-drop push-ups1), 7.6 cm (box-drop push-ups2) and 11.4 cm (box-drop push-ups3) variants are more intense compared to the other analysed push-up variations. Push-ups from kneeling positions and with elevated hands were less intense compared to the other analysed push-up variations (Table 5). Moore et al. (2012) examined the kinetic data related to box-drop push-ups1, box-drop push-ups2, box-drop push-ups3 and clap push-ups. The *PF* was significantly greater for the clap push-ups than for all

other variations (Garcia-Masso et al., 2011). However,Hinshaw et al. (2018) quantified the changes in peak force, peak power and peak velocity for a modified plyometric push-up on the knees and for plyometric push-ups with or without external loading (5% or 10% of *BW*) in physically active young adults. The main finding of his study revealed that the peak force and the force at peak velocity were higher for the push-up without external loading compared to the two types of push-ups with external loading. The same authors (Hinshaw et al., 2018) suggested that push-ups without external loading may be more beneficial when a high movement speed is desirable and push-ups with external loading may be preferable when greater force production is required. The latter results were also supported by Wang et al. (2017), who investigated ground reaction force patterns during ballistic push-ups (i.e. countermovement push-ups) in 60 recreationally active men, and found peak forces ranging from 120% to 121% *BW*.

It is not surprising that the forces associated with plyometric push-ups were lower than for plyometric exercises involving the lower limbs, since the latter involves the entire *BW*, while the upper-limbs only support part of the *BW*. The plyometric push-ups impose a load of about three-quarters of the *BW* (Dhahbi et al., 2017a; Koch et al., 2012). Thus, it is imperative that strength and conditioning coaches understand that the *PF* in relation to the musculoskeletal system of each upper extremity was 0.7 times that of the *BW* (a total of 1.4 times that of the *BW* across both extremities) (Dhahbi et al., 2017a; Koch et al., 2012). This finding can be used when calculating the total loading athletes have to withstand during a workout, in order to ensure sufficient stimulus for adaptation while avoiding overloading risk (Dhahbi et al., 2017a; Koch et al., 2012).

Impact force

The studies conducted by Lo, McCabe, DeGoede, Okuizumi, and Ashton-Miller (2003) and Garcia-Masso et al. (2011) indicate impact-force values of approximately 700 N (i.e. 100% *BW*) when performing plyometric push-ups. Dhahbi et al. (2017a) revealed that the Impact force was higher in standard countermovement push-ups and standard squat push-ups, compared to the other exercises. For standard countermovement push-up and standard squat push-up positions, the unsupported body mass is greater, therefore the resulting force application is greater. Moreover, the impact-force values ranged from 136% to 192% *BW* for all exercises. Several authors (Garcia-Masso et al., 2011; Koch et al., 2012; Moore et al., 2012; Schulte-Edelmann et al., 2005) demonstrated that the Impact force was influenced not only by the height of fall but also by the neuromuscular strategy applied during the landing phase. To this extent, it seems that the impact-force intensities must be taken into consideration because they are influential and could be adapted to any situation in order to increase or avoid overload. So far, the jump push-up and fall push-up exercises are not considered inadvisable for young people. Therefore, young people can use these two push-up variants to achieve greater training-related intensity.

Compared to other plyometric push-up exercises, the fall push-up exercises may have a greater effect on the musculoskeletal and nervous systems (i.e. the upper-limb adaptation to the fall and the damping reflex on impact) (Garcia-Masso et al., 2011). Therefore, this form of exercise should be used with individuals who possess greater physical fitness. Alternatively, for participants who aim to develop power through a plyometric method without undergoing a large Impact force, the drop-fall push-up is

recommended. During these stages, lower-intensity loads should be applied with the purpose of adequately preparing the musculoskeletal system for subsequent stress (Garcia-Masso et al., 2011). Koch et al. (2012) highlighted an observed asymmetry between the upper-limbs while performing push-ups. During the impact, the dominant limb undergoes/produces the highest Impact forces. Future research could reveal the underlying aetiology of this asymmetry.

Following the law of free fall, a fall from a greater height would result in greater impact forces. In addition, the Impact forces are absorbed by the passive structures of the musculoskeletal system (Chou et al., 2001; Kim & Ashton-Miller, 2003), which may lead to osteogenic adaptations or potential injury (Burgess, Connick, Graham-Smith, & Pearson, 2007; Daly, Rich, Klein, & Bass, 1999; Ebben, Fauth, Kaufmann, & Petushek, 2010). The results obtained by Moore et al. (2012) confirm the importance of quantifying Impact forces while carrying out plyometric exercises involving the upper-limbs. It is noteworthy that an impact-force intensity equal to twice that of the *BW* would be considered low for lower-extremity plyometric activities, whereas for the upper-limbs the same intensity could be considered medium to high (Nigg, Cole, & Brüggemann, 1995; Santos-Rocha, Veloso, & Machado, 2009).

The rate of force development (RFD)

To our knowledge, only five studies (Dhahbi et al., 2017a; Garcia-Masso et al., 2011; Hogarth et al., 2013; Hrysomallis & Kidgell, 2001; Koch et al., 2012) have investigated the *RFD* in the kinetic analysis of push-ups. Hogarth et al. (2013) examined the test–retest reliability of *GRF*-based parameters collected from standard countermovement push-ups. Hrysomallis and Kidgell (2001) used the positive maximum *RFD*, derived from an explosive push-up, as a power index of the upper-limbs (Table 6). Dhahbi et al. (2017a) similarly found that the propulsion *RFD* values are larger in countermovement push-up exercises at 2.28 *BW*/s, compared to squat push-up exercises at 0.57 *BW*/s. In addition, Wang et al. (2017) also examined the propulsion *RFD* of ballistic push-ups and reported values of 5.89–5.99 *BW*/s. The authors (Dhahbi et al., 2017a; Wang et al., 2017) explained these results by noting that the technique of the countermovement push-up uses the stretch-shortening cycle to store and release elastic energy, which may produce a force more quickly (Duchateau & Baudry, 2014; Earp et al., 2011) rather than just using a concentric-type muscle contraction (e.g. squat push-ups). Meanwhile, as no significant differences were found between all exercises for the *RFD* of impact, these findings disagree with the results of Garcia-Masso et al. (2011), who found that the *RFD* of impact was larger in fall push-ups than in jump push-ups. They (Dhahbi et al., 2017a) attributed this observation to altered landing strategies adopted by the participants (Koch et al., 2012). Koch et al. (2012) compared the *RFD* of impact between the dominant and non-dominant upper-limb in four plyometric push-up variations (clap push-up, 3.8-cm box-drop push-up, 7.6-cm box-drop push-up and 11.4-cm box-drop push-up). These authors reported a higher *RFD* of impact in the dominant limb than in the non-dominant limb in all four push-up variations. Furthermore, for both limbs, the clap *RFD* of impact was significantly greater than that for box-drop push-ups1, box-drop push-ups2 and box-drop push-ups3. The clap form of push-ups demonstrated the highest *RFD* of impact, indicating that force is applied more quickly than inbox-drop push-ups1, box-drop push-ups2 and box-drop push-ups3, with little difference

Table 6. Push-up rate of ground reaction forces expressed as a coefficient of total body mass ground reaction force per second.

	Rate of impact force (*BW/s*)	Propulsion rate force (*BW/s*)
Fall push-ups (Garcia-Masso et al., 2011)	27.79 ± 3.08	1.80 ± 0.10
Jump push-ups (Garcia-Masso et al., 2011)	16.82 ± 1.79	1.53 ± 0.14
Countermovement push-ups (Garcia-Masso et al., 2011)	-	2.08 ± 0.13
Clap push-up (Koch et al., 2012)	7.26 ± 0.20*	-3.32 ± 0.14*
3.8cm box-drop push-up (Koch et al., 2012)	4.22 ± 0.24*	-2.87 ± 0.12*
7.6cm box-drop push-up (Koch et al., 2012)	4.92 ± 0.28*	-2.99 ± 0.11*
11.4cm box-drop push-up (Koch et al., 2012)	6.27 ± 0.42*	-2.87 ± 0.12*
Explosive push-up (Hrysomallis & Kidgell, 2001)	-	5.88 ± 1.23
Standard countermovement push-up (Wang et al., 2017)	-	5.99 ± 2.35
Standard countermovement push-up (Dhahbi et al. 2017a)	48.51 ± 38.72	2.28 ± 0.84
Standard countermovement push-up (Hogarth et al., 2013)	-	7.65 ± 2.23
Standard squat push-up (Dhahbi et al. 2017a)	57.37 ± 66.79	0.57 ± 0.59
Kneeling countermovement push-up (Dhahbi et al. 2017a)	36.46 ± 26.33	1.55 ± 0.59
Kneelingsquat push-up (Dhahbi et al. 2017a)	39.45 ± 32.27	0.32 ± 0.16
Drop-fall push-up (Dhahbi et al. 2017a)	41.47 ± 32.54	3.02 ± 1.50

*Dominant limb; *BW* = body weight

observed between the *PFs* in these push-ups. Based on the significantly higher propulsion RFD, it appears that the clap push-up is the most intense form of push-up. Furthermore, Koch et al. (2012) recommended that the increase in the height of box-drop push-ups is an adequate stimulus for progressively increasing exercise demands.

Suspended push-ups

F_0

The study by Gulmez (Gulmez, 2016) determined and compared the loads at four different angles during TRX push-ups (which consist of push-ups holding TRX straps which hang vertically). This research revealed that by decreasing the TRX angle and/or when the elbows were in flexion, the elbow loading steadily increased. Gulmez (2016) further reported that when the TRX angle was set at 0° and the elbows were extended, 50.4% *BW* was recorded. However, when the elbows were at maximum flexion, 75.3% *BW* was supported on the straps. There was a more than four-fold increase in the loading on the upper extremities when the TRX angle was changed from 45° to 0°. These results agree with a recent study (Giancotti et al., 2018) which showed that when the length of the TRX device increased, the body inclination decreased, while the ground reaction force at the feet decreased and the load supported by the arms increased. Moreover, when the participant moved from extension to flexion of the elbow, the load on the TRX device increased. Based on this finding, the TRX angle may be reduced to facilitate progression of training intensity. Furthermore, the instability of the TRX push-up must also be taken into account as a factor affecting the relative load at various angles.

Vertebral-joint compressive forces/loads

The proper performance of push-ups requires the performer to maintain a static and standard lumbar spine posture while maintaining a balance of musculoskeletal and joint systems against the force of gravity (Beach et al., 2008). The osteoligamentous structures of the spine are less resistant to bending movements from the standard back

posture (Panjabi, 1992). McGill et al. (2014) and Beach et al. (2008) kinetically analysed the differences between stable and suspended push-up exercises, evaluating the spine compressive loads. They showed (Beach et al., 2008; McGill et al., 2014) that there were different levels of spine load for each push-up variation. In addition, there was greater muscle activity associated with labile surface exercises to maintain the correct position of the body compared to exercises on stable surfaces (Beach et al., 2008). However, suspended push-ups resulted in more intense spine loads than push-ups on a stable surface (Beach et al., 2008). Therefore, the decision to use suspended push-ups should take into consideration the history of spine injury, training targets and the fitness level of the performer (Beach et al., 2008; McGill et al., 2014). Labile push-ups elicit significantly greater L4/L5 compression (Beach et al., 2008; McGill et al., 2014). All pushing exercises analysed in the study by McGill et al. (2014) produced spine compressions of less than 2 kN. On the other hand, in the study of Beach et al. (2008), some suspended push-up variations (e.g. TRX) produced peak L4/L5 intervertebral-joint compressive force loads of around 3.4 kN, exceeding the reference limit value established by the *National Institute of Occupational Safety and Health* (NIOSH, 1981). The *NIOSH* limit value was fixed to ensure the safety of the spine. Another study (Genaidy, Waly, Khalil, & Hidalgo, 1993) suggested that the spine osteoligamentous structures of boys can resist compressive loads of more than 17 kN without any injury risk.

Limitations

The first limitation of the current review is the fact that we were unable to evaluate the reliability, validity and sensitivity of each kinetic parameter used in the included studies. In addition, some studies tended to calculate the mechanical power outputs of explosive push-ups, and their results (Hinshaw et al., 2018; Speranza et al., 2018; Wang et al., 2017) should be regarded with caution, as the calculation methods used to find mechanical power outputs are open to criticism (Dhahbi, Chaouachi, Cochrane, Chèze, and Chamari 2017b; Wang et al., 2017). Thus, assessing the validity of the method of estimating power output when performing push-ups, remains a point in need of future attention.

Conclusions

In the current review, specific kinetic details are gathered for each push-up variation. Based on the reviewed studies, the F_0 with the elbows flexed is significantly greater (8–20% *BW*) than that with the elbows extended. The load supported by one limb ranged from 6% *BW* for prayer push-ups to 60% *BW* for one-arm push-ups. Regarding the *PF* parameter, the clap push-up imposed the highest exercise intensity. Plyometric push-ups achieved a higher propulsion rate force without causing very high impact forces (i.e. they did not exceed forces of twice the body weight). Fast push-ups and push-ups with the shoulder adducted resulted in the highest peak elbow flexion moments. Suspended push-ups resulted in the highest vertebral-joint compressive forces. Except for suspended push-up variations, the lateral and medial elbow-joint force, the peak force and rate of impact force are greater on the dominant limb than on the non-dominant limb. Moreover, outcomes on the effects of sex on push-up kinetic parameters are conflicting.

The push-ups with less wrist and elbow-joint stress are the ones that adopt a neutral hand position.

The kinetic data detailed in this review for the various push-up exercises could be used to individualise training load as a percentage of *BW* in an upper-body resistance-training programme (Dhahbi et al., 2017a). The current review showed that the intensity of push-up exercises can be adjusted by altering the starting position (e.g. feet or knees, hand position, feet and/or hand height) and by altering the execution speed, the surface type and the *ROM*. It is imperative to determine the client's or athlete's fitness level, the goals of the training and the appropriate training load stimulus, in order to select the appropriate push-up exercise.

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ORCID

Johnny Padulo http://orcid.org/0000-0002-4254-3105

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